



SUSHANTH PAATNAIK

INVENTOR · DEEP-TECH · FOUNDER

Engineering
Matter.

Engineering
Capital.

Engineering
Scale.

From a maker's bench to global impact. This journey is about engineering breakthrough materials and ventures that build a more sustainable, abundant and equitable world.



ABOUT

SUSHANTH PAATNAIK

INVENTOR • DEEP-TECH FOUNDER • INDUSTRY BUILDER

Building industrial systems
that outlive the inventor.

Sushanth Paatnaik is an inventor, deep-tech entrepreneur and industry builder focused on transforming breakthrough science into globally scalable industrial solutions. His work spans advanced materials, nanotechnology, clean energy and sustainable manufacturing, with multiple innovations recognized nationally and internationally.



PROFILE

Inventor
Deep-Tech Founder
Industrial Strategist
Speaker



ORIGIN

Bhubaneswar, Odisha



WORK BASE

Ahmedabad
New Delhi
Bhubaneswar



EDUCATION

B.Sc. @ IISER Bhopal
B.E. (EC) @ OCT Bhopal

FOCUS AREAS



Advanced
Materials



Nanotechnology
& Graphene



Industrial
Innovation



Sustainable
Manufacturing

“

Engineering intelligent materials and scalable technologies that create lasting industrial impact.

”

INVENTING • BUILDING • IMPACTING

ORIGIN

Every idea has a birthplace.



Childhood
curiosity in
a small town



First workshop
at age 14



Breath-powered
wheelchair
that started it all



PHILOSOPHY

The future
is not
imagined.
It is
engineered
quietly.



EARLY WORKS

The teenage innovations that earned six Presidential Awards.

Between 2008 and 2013 — still in his teens — Sushanth designed and built six original devices, each tackling a real human problem with hardware that worked.



BREATH-POWERED WHEELCHAIR

2008

Developed India's first wheelchair operated using human breath. Recognized by the President of India and became the foundation of my innovation journey.



ACCIDENT-PROOF RETROFIT KIT

2011

A retrofittable safety kit designed to transform any existing vehicle into a safer machine — without redesign of the host platform.



SHE WATCH

2012

A discreet, smart-watch form factor wearable engineered to alert and locate — designed long before the category became mainstream.



SOL-KA

2012

Ambient-light charging power bank that harvests indoor light to keep devices topped up — a quietly self-sufficient piece of everyday hardware.



SUPER SENSE TECHNOLOGY

2013

Gesture-controlled computing enabling computers to be operated through hand gestures alone — exploring a natural human-machine interface.



HIGH-EFFICIENCY POWER KIT

2008

A high-efficiency, low-cost power generation kit conceived to bring reliable electricity to remote rural communities.

“

*Every idea was a small step towards solving a real problem.
These early innovations shaped the path ahead.*

”

Sushanth Paatnaik

SUSHANTHPAATNAIK.COM

JOURNEY

◆ A TIMELINE OF INNOVATION, IMPACT & INDUSTRY CREATION ◆

From a young inventor to a deep-tech entrepreneur,
building technologies and companies that solve real-world problems.



2008

FIRST PRESIDENTIAL AWARD



2010

INK FELLOW



2012

NASA AWARDEE



2015

INTHINKS FOUNDED



2020

CAPATTERY



2025

MONOATOM LABS



TODAY

GLOBAL DEEP-TECH ECOSYSTEM

2010

MIT TR35



2012

TED SPEAKER



2013

SIX PRESIDENTIAL AWARDS



2016

ZZEN-X INVERTER LED



2024

SPI INDUSTRIES



2025

GRAFILLIUM



“

Every idea was a small step towards solving a real problem.
These early innovations shaped the path ahead.

”

Sushanth Paatnaik

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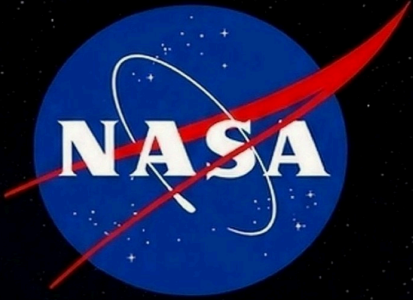
HONORS & RECOGNITIONS



**PRESIDENT
PRESIDENT
OF INDIA**

AWARDS X 6

Honored by the President of India for six consecutive years for outstanding innovation and leadership.



**NASA
AWARDEE**

Recognized by NASA for contributions in innovation and technology.

**MIT
TR35**

**MIT TECHNOLOGY
REVIEW 35 UNDER 35**

Recognized among the world's top young innovators under 35.

**TED
SPEAKER**

Invited TED Speaker sharing ideas that inspire and create impact.



intel.

INTEL INNOVATOR

Recognized by Intel for driving breakthrough technologies.

INK

INK FELLOW

Honored by India Innovation Network for exceptional innovation and impact.

NIF

NIF AWARDEE

Recognized by the National Innovation Foundation for grassroots innovation.

HALL OF FAME

RECOGNITIONS • AWARDS • HONORS

A journey of innovation, impact and inspiring a better tomorrow.



2009
National Recognition



2010
Youth Leadership



2013
Innovation Champion

◆ **PRESIDENT OF INDIA**
SIX-TIME NATIONAL AWARDEE
2008 • 2009 • 2010 • 2011 • 2012 • 2013

India's Highest National Recognition
for Young Innovators



NASA Award
2011



MIT TR35 Innovator
2011



Innovation Showcase
2010



Fab-10 Barcelona
2013



Fab-11 MIT Boston
2014



INK Fellow
2010



Intel IRIS Award
2010



Young Achiever
2013



STPI Chunaui
2021



Elecrama Recognition
2024



CEO Clubs Leadership
2025



BRICS Global Delegate
2023



FICCI Bharat Summit
2024



Manufacturing Leadership
2023



Industry & Innovation Excellence
2025



Mobility Innovation
2019



Social Impact Recognition
2018



Youth Icon Award
2016



Entrepreneurship Excellence
2022



National Innovation Recognition
2017

A LEGACY OF INNOVATION

2008
Journey
Begins

2010
Early
Recognitions

2012
National
Honors

2013
Global
Milestones

2016
Expanding
Impact

2018
Inspiring
Change

2021
Building
The Future

2025
Continuing
The Mission

INNOVATOR • ENTREPRENEUR • SPEAKER
ADVISOR • MENTOR • DEEP-TECH BUILDER

Sushanth Paatnaik
SUSHANTHPAATNAIK.COM

BUILDING INTELLIGENT MATTER.
TRANSFORMING INDUSTRIES.
IMPACTING BILLIONS.

BY THE NUMBERS

A track record of innovation, impact
and global recognition.

15+

—
YEARS

10+

—
INDUSTRIAL
TECHNOLOGIES

6

—
PRESIDENTIAL
AWARDS

5

—
VENTURES

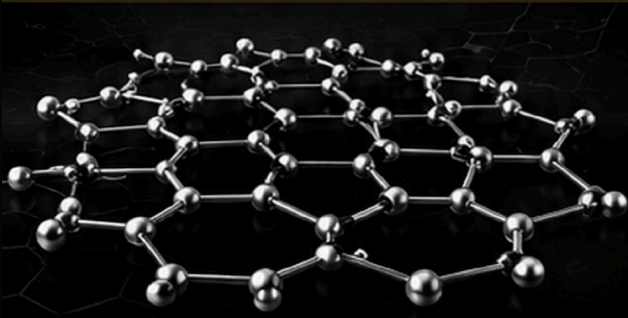
50+

—
MEDIA
FEATURES

100+

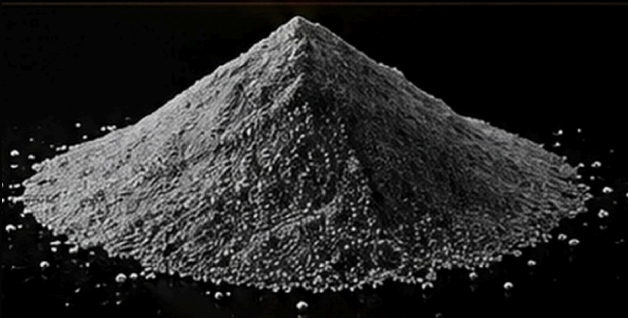
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INDUSTRY
COLLABORATIONS

ENGINEERING INTELLIGENT MATTER



GRAPHENE

The foundation of next-generation materials. Stronger, lighter, smarter by design.



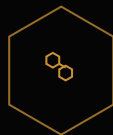
ADDITIVES

Nano-engineered additives that enhance performance, efficiency, and sustainability.



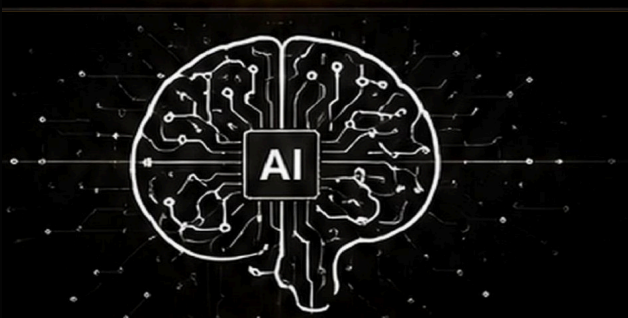
COATINGS

Advanced functional coatings for protection, performance, and precision.



COMPOSITES

High-performance composites engineered for tomorrow's challenges.



AI

Intelligent systems and data-driven innovation powering the future of materials.

23

INNOVATIONS ENGINEERED FOR IMPACT

ADVANCED MATERIALS

CLEAN ENERGY

INDUSTRIAL SOLUTIONS

“
Engineering intelligent materials and
scalable technologies that create lasting
industrial impact.”

23
INNOVATIONS
Across Diverse Industries

3
DEVELOPMENT
STAGES
Commercial • Pilot • R&D

SUSTAINABLE IMPACT
Building technologies for a
better, stronger and cleaner
planet

STAGE I – COMMERCIAL / FIELD DEPLOYED (1–6)

1 GRAPHACRETE Graphene Concrete Admixture



CONSTRUCTION

- ↑ 30–45% Strength
- ↓ 15–25% Cement Use
- ↓ 20–30% CO₂ Emissions

2 GRAFFISOL Graphene Solar Coating



SOLAR ENERGY

- ↑ 5–12% Solar Efficiency
- ↓ 10–18°C Surface Temperature
- ↑ 20% Panel Life

3 CERAPHENE Graphene Ceramic Coating (Car Body)



CAR PROTECTION

- ↑ 3x Hardness
- ↑ 60% Wear
- ↓ 5x Corrosion Resistance

4 HD-G-PE Graphene HDPE Masterbatch



POLYMERS

- ↑ 30–40% Tensile Strength
- ↓ 15% Material Usage
- ↑ 2x Service Life

5 GRAPHENODES Graphene Electrode Materials



BATTERIES

- ↑ 25–40% Energy Density
- ↑ 3x Charging Speed
- ↑ 2x Cycle Life

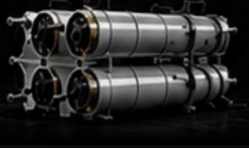
6 IGNITRON D Diesel Combustion Optimization



MOBILITY

- ↑ 8–15% Fuel Efficiency
- ↓ 35–60% Smoke
- ↓ 20–30% Emissions (NO_x/CO)

STAGE II PLANT & FIELD PILOT (7–16)

7 COALORIX Coal Combustion Optimization  THERMAL POWER ↑ 5–10% Coal Consumption ↓ 3–6% Boiler Efficiency ↓ 15–30% Fly Ash	8 AQUAMAX High Voltage, Low Current Water Recovery System  WATER ↑ 30–50% Water Recovery ↑ 15–20% Water Reuse ↑ 15–20% Cooling Efficiency	9 IGNITRON P Petrol Combustion Optimization  MOBILITY ↑ 6–12% Mileage ↓ 20–30% Emissions	10 LUBRITRON Nano Engine Oil Additive  LUBRICANTS ↓ 40% Engine Wear ↑ 2x Oil Life ↑ 3–6% Fuel Economy	11 RUSTENE Anti-Corrosion Coating  COATINGS ↑ 5x Corrosion Protection ↑ 3x Coating Life
12 GRYOGEN Hydrogen Separation Membrane  H₂ HYDROGEN ↑ 99.9%+ Hydrogen Purity ↓ 30% Separation Energy	13 MARIPHENE Desalination Membrane  WATER ↓ 20–35% Energy Consumption ↑ 30% Water Recovery	14 AEROPHENTER Atmospheric Water Harvesting  WATER Produces 5–25 L/day of Water per Unit* (Climate Dependent)	15 FIBRASHENE Reinforced Graphene Fibres  COMPOSITES ↑ 40–60% Tensile Strength ↓ 25% Weight	16 VOLTAPHENE Graphene Battery Cell  ENERGY STORAGE ↑ 2x Lifespan ↑ 3x Fast Charging Capability ↑ 30% Energy Density

STAGE III R&D / BENCH (17–23)

17 ARMOPHENE Graphene Armour  DEFENCE ↑ 30–40% Weight Enhanced Ballistic Resistance	18 HYDROCELL Fuel Cell Technology  H₂ HYDROGEN ↑ 15–20% Fuel Cell Efficiency ↓ Catalyst Usage	19 BITUMAX Bitumen Additive  INFRASTRUCTURE ↑ 2x Pavement Life ↓ 40–60% Rutting & Cracking	20 PYRONEX Fire & Thermal Protective Additive  COATINGS ↑ Fire Resistance up to 1200°C ↑ 30% Thermal Insulation	21 GRAPHYRE Graphene Tyres  MOBILITY ↑ 40–60% Tyre Life ↓ 15% Rolling Resistance	22 GRAPHOSITE Structural Graphene Composites  COMPOSITES ↑ 35–50% Structural Strength ↓ 25–40% Weight	23 THERMAPHENE Temperature-Regulating Textile  SMART MATERIALS ↑ 5–8°C Passive Temperature Reduction ↑ Thermal Comfort
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SUSTAINABLE

Solutions that reduce emissions, conserve resources and protect our planet.



SCALABLE

Technologies engineered for scale and real-world industrial adoption.



IMPACTFUL

Delivering measurable performance, efficiency and value.



GLOBAL VISION

Innovations for industries and communities worldwide.

Portable RO with Reject-Water Recovery

Conventional portable RO discards three litres for every one it purifies. This proposal pairs a closed-circuit recirculation loop with a graphene-oxide recovery cartridge to reclaim a significant share of it — without giving up the compact footprint.

THE PROBLEM

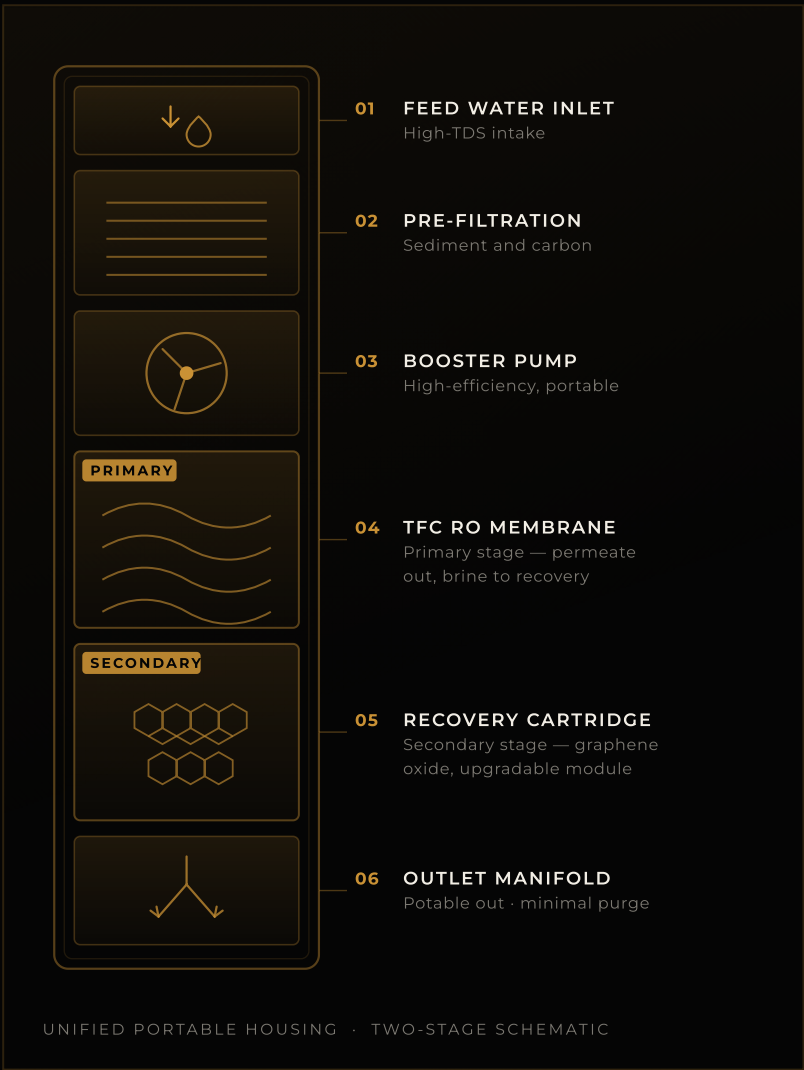
RO forces water through a semi-permeable membrane with pore sizes near 0.0001 microns. It is highly effective at reducing Total Dissolved Solids and removing heavy metals, but standard residential and portable units run a 3:1 waste ratio — 75% of intake is discarded to stop mineral crystallisation scaling the membrane.

THE ARCHITECTURE

Primary stage. A high-efficiency booster pump and a Thin Film Composite membrane produce potable water, separating salts and hardness minerals into a concentrated brine stream.

Secondary stage. Rather than routing brine to a drain, it is diverted into a recovery attachment built on advanced materials and modified flow dynamics — because standard TFC membranes cannot survive the osmotic pressure or the rapid fouling that concentrated brine causes.

A next-generation portable purifier that processes its own reject water — significantly increasing the overall recovery rate while maintaining a compact footprint.



3:1

CONVENTIONAL
WASTE RATIO

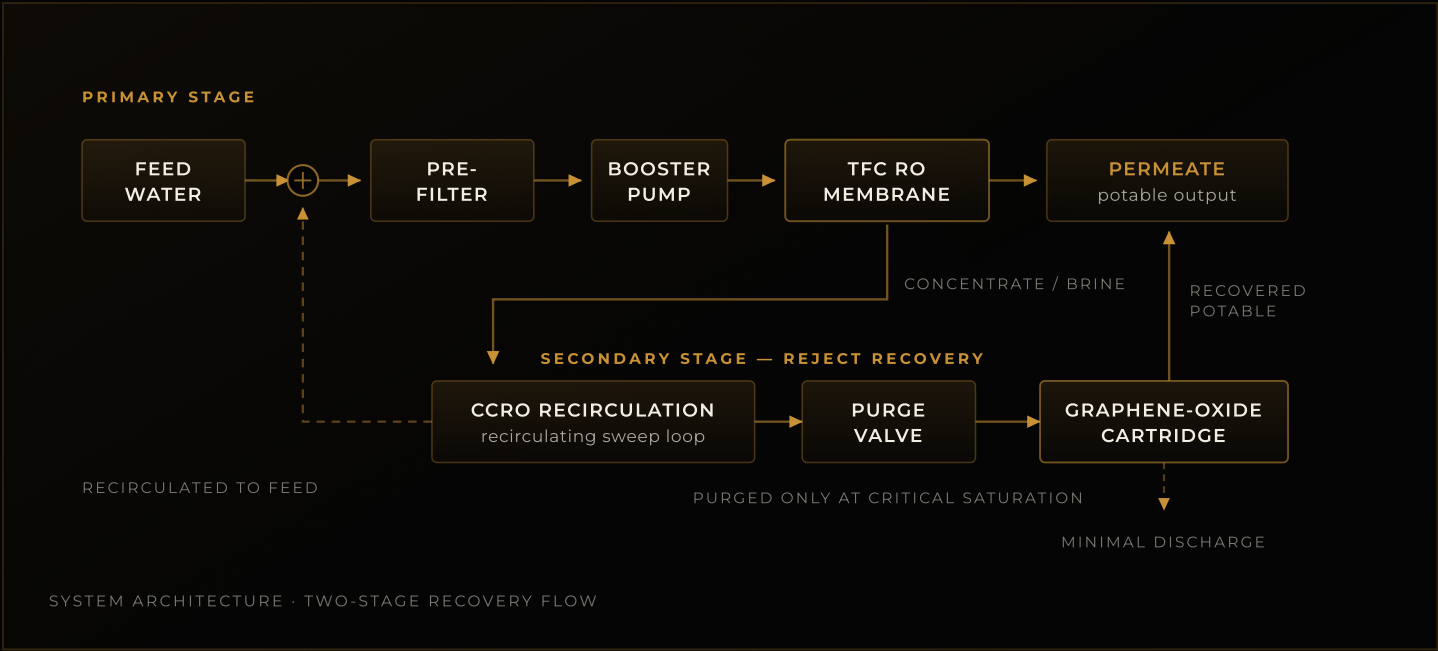
0.0001

MICRON MEMBRANE
PORE SIZE

75%

OF INTAKE DISCARDED
BY STANDARD UNITS

Closed-circuit recirculation, graphene-oxide recovery.



ECONOMIC FEASIBILITY FOR PORTABLE UNITS

TECHNIQUE	CAPEX / OPEX	COMMERCIALISATION ASSESSMENT
Graphene Oxide (GO) Membranes	Moderate–High CAPEX Low OPEX	High initial setup cost — scaling uniform nanomaterial manufacturing is hard. Best portable performance: low pressure requirement, high fouling resistance. Ideal for premium positioning.
Miniaturised CCRO	Low CAPEX Moderate OPEX	Highly economical. Uses existing pump architecture plus smart valves and sensors. Excellent candidate for immediate commercialisation.
Electrodialysis Reversal (EDR)	High CAPEX Low OPEX	Miniature electrodes and ion-exchange membranes remain prohibitive for consumer portables, but viable for specialised military or emergency-relief systems.
Thermal Evaporation (ZLD)	Highly uneconomical	The energy required to boil brine is massive. Completely unsuited to portable, battery-operated or low-power applications.

STRATEGIC RECOMMENDATION

A hybrid approach. The primary stage runs a standard TFC membrane with a CCRO recirculation loop to cut waste economically; the secondary reject attachment becomes an upgradable graphene-oxide cartridge, targeting the premium segment where higher CAPEX is justified by maximum water recovery in off-grid scenarios.

INDUSTRIAL TRANSLATION

TRANSFORMING INNOVATION INTO REAL-WORLD IMPACT



SOLAR

Advanced coatings for higher efficiency and durability.



ENERGY

Materials and technologies driving cleaner, smarter energy solutions.



AUTOMOTIVE

High-performance materials for mobility and efficiency.



INFRASTRUCTURE

Stronger, smarter solutions for resilient and sustainable infrastructure.



DEFENSE

Advanced composites for protection, performance, and mission criticality.



CONSTRUCTION

Innovative materials for stronger, longer-lasting and sustainable builds.



VENTURES



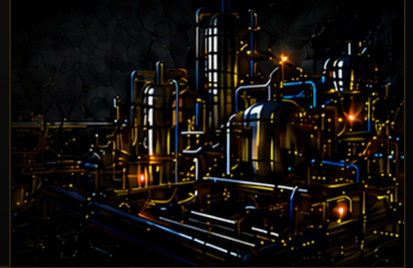
SPI INDUSTRIES
R&D INDUSTRIAL SOLUTIONS

● SPI INDUSTRIES

R&D Industrial Solutions

FOUNDER

Driving industrial innovation through advanced R&D and engineering solutions.



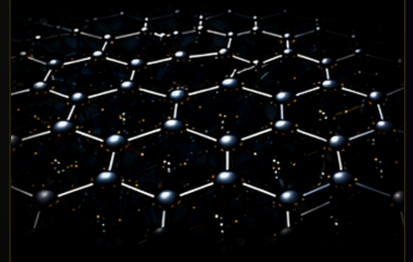
MONOATOM
LABS

● MONOATOM LABS

Graphene at Scale

CO-FOUNDER & CEO

Pioneering graphene production at scale and developing innovative applications.



GRAFFILLIUM
DEEP TECH
INNOVATIONS

● GRAFFILIUM

Eco-Nano Additives

CO-FOUNDER & CIO

Developing eco-nano additive solutions for industrial and energy applications.



INTHINKS
INNOVATION STUDIO

● INTHINKS

Innovation Studio

FOUNDER

Deep-tech research and product development for real-world challenges.



STARUNICO
INVESTMENT

● STARUNICO CAPITAL

Deep-Tech Investment

FOUNDER

Investing in breakthrough deep-tech startups building a sustainable and intelligent future.



MAGPPIE

● MAGPPIE

Stone Wellness Kitchens

CHIEF INNOVATION OFFICER (CIO)

Pioneering the world's first 100% stone-built modular kitchen solutions.



◆ COUNSEL ACROSS DIVERSE INDUSTRIES. ◆

A short ledger of the houses I quietly advise — one mark per layer of the industrial network.

VIIIROX

V VPRPL

WEHEAR

Tileopedia

EUROGRES
Ceramics that Europees

GREENOMERS
GREEN POLYMERS

NEWS & MEDIA

A LEGACY DOCUMENTED
BY THE WORLD'S LEADING PUBLICATIONS.

From national headlines to global platforms,
the journey of innovation, impact and
entrepreneurship continues to inspire.



16+
YEARS OF
MEDIA
PRESENCE

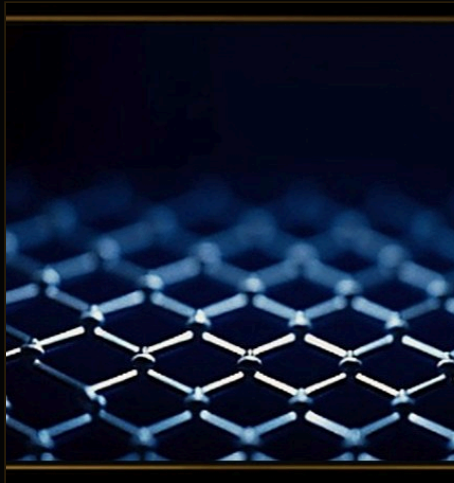
50+
NATIONAL &
INTERNATIONAL
FEATURES

COVERAGE
ACROSS
**NATIONAL &
GLOBAL
PLATFORMS**

FEATURED COVERAGE



Six times President awardee profile
The Global Indian | 2013



GraphIN 2026 graphene coverage
Rediff / PTI | 2026



Battery charging innovation
Business Standard | 2024



Serial entrepreneur profile
India Today | 2016



Patent run and awards
Times of India | 2012



Capattery graphene energy storage
YourStory | 2023

FEATURED IN

**INDIA
TODAY**

**THE TIMES
OF INDIA**

Business Standard

ThePrint

**DECCAN
Chronicle**

**The Telegraph
online**

**THE GLOBAL
INDIAN**

**MIT
Technology
Review**

**TED
India**

National Innovation Foundation

GOVERNANCE now

**rediff.com
PTI**

YOURSTORY

**PRODUCT
NATION**

WeRIndia

**THE NEW
INDIAN EXPRESS**

**THE NEW
INDIAN EXPRESS**

“

From local headlines to global platforms,
the journey of innovation continues to inspire.

”

SPEAKING ENGAGEMENTS

From thought leadership talks to global conferences, sharing insights on innovation, deep-tech, and building a sustainable future.



TED SPEAKER



Delivered a thought-provoking talk on innovation and the future of technology at a TEDx event.



CEO CLUBS & LEADERSHIP FORUMS



Engaged with industry leaders and CXOs to discuss strategy, growth, and the future of deep-tech ventures.



UNIVERSITIES & INSTITUTES



Invited by leading universities to speak on entrepreneurship, deep-tech innovation, and building impactful solutions.



GLOBAL CONFERENCES



Participated in global platforms to share insights on emerging technologies and sustainable innovation.



SILICON VALLEY & BEYOND



Connected with innovators and investors across Silicon Valley and beyond to explore the future of deep-tech.

VOICES

ENDORSEMENTS & TESTIMONIALS

What people say about the work.

On-record testimonials from innovation foundations, global press, industry leaders, and international stages.



06

PRESIDENTIAL
AWARDS



20+

INSTITUTIONAL
VOICES



40+

EDITORIAL
FEATURES



GLOBAL

STAGES & OUTLETS



National
Innovation
Foundation

Accelerating Inclusive Innovations

“

Sushant is an amazing innovator and always innovates with high social impact.

– Anil K. Gupta

Professor, IIM-Ahmedabad
Vice-Chair



“

'A life-changing innovation 'Enabler' for the disabled — highly appreciable.'

– Mr. Srinivas

Senior Journalist

Deloitte.

“

The best AI live demo ever seen so far. This demo really made my day.

– P. R. Ramesh

Chairman



IndianOil

“

Very effective and innovative solution for underground pipelines.

– Mr. Joseph

Eastern Zone



“

A very inspiring entrepreneur and great social-revolutionary products.

– Shradha Sharma

Founder & CEO

”

COLLABORATION



INDUSTRY

Co-creating
impact at scale



INVESTMENT

Funding deep-tech
moonshots



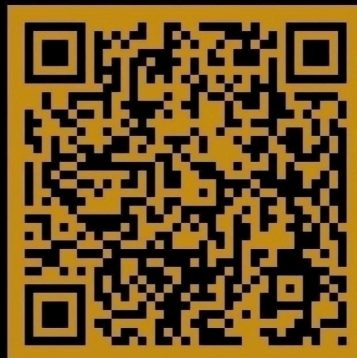
RESEARCH

Pushing the
boundaries
of science



GOVERNMENT

Building systems
for the future



Let's build the future together.

sushanthpaatnaik.com

Building
industrial
systems
that outlive
the inventor.



SUSHANTH PAATNAIK



sushanthpaatnaik.com



info@sushanthpaatnaik.com



Ahmedabad | New Delhi | Bhubaneswar

